OcuHLE-X

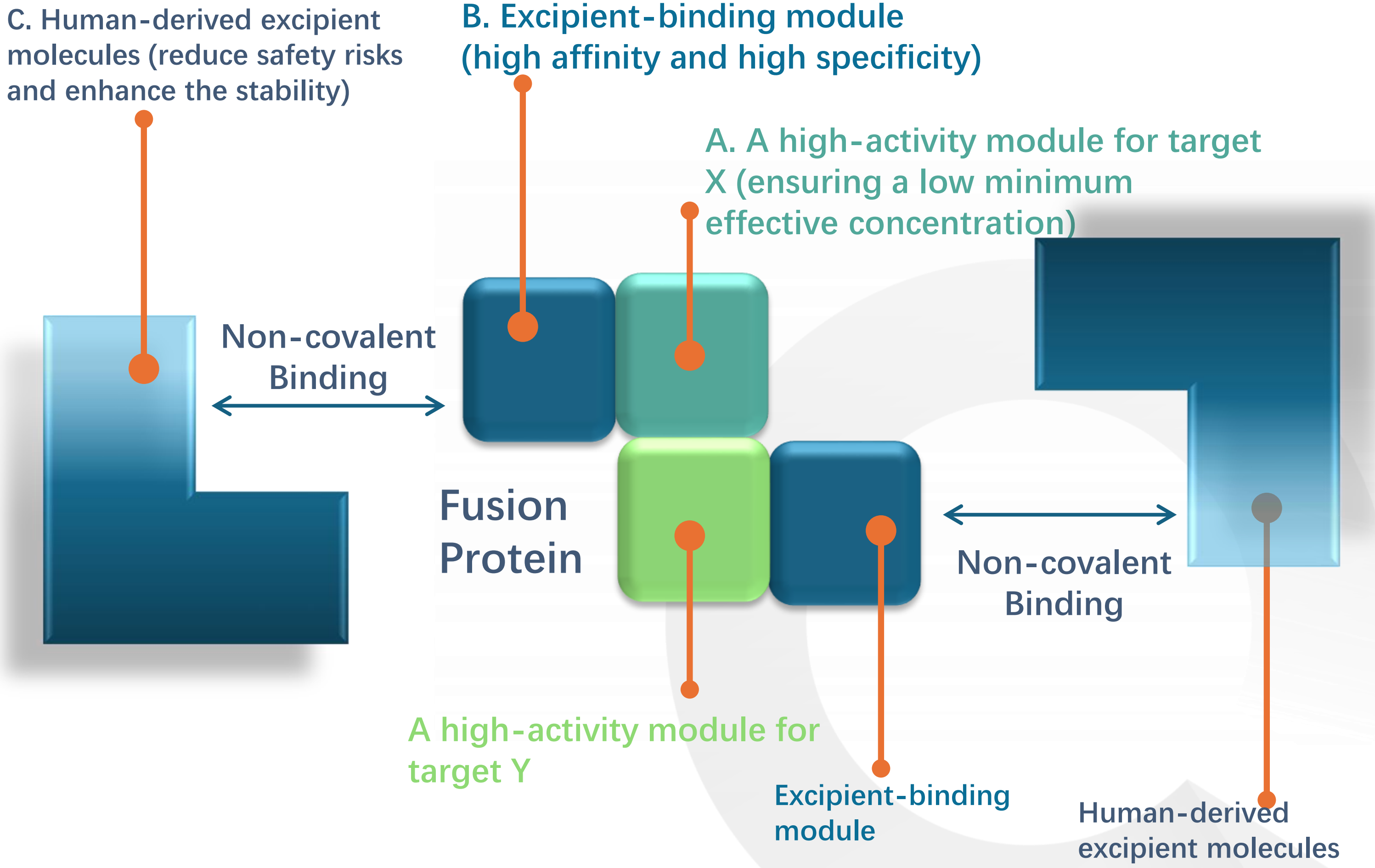
Platform

OcuHLE-X® Design Scheme

"Molecular + Formulation Engineering" Holistic Solution

Platform Features:

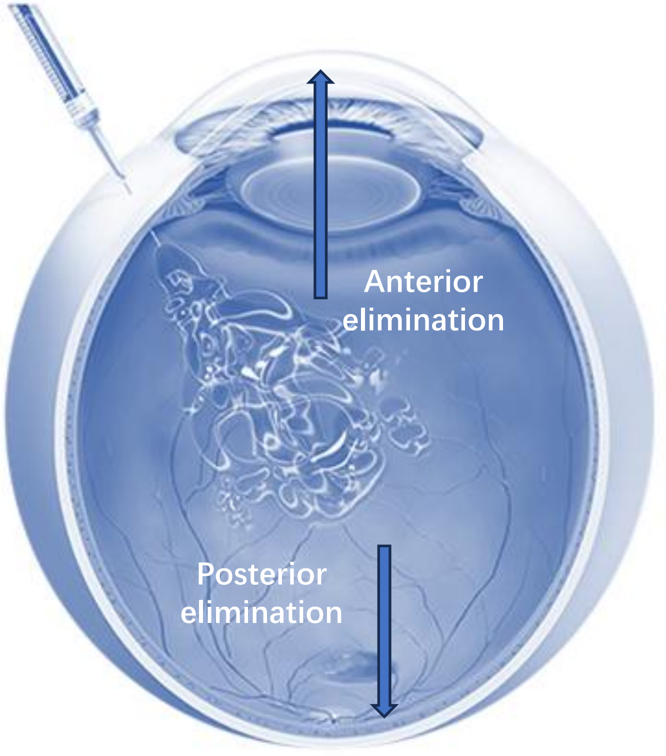
- Maintains low viscosity and high stability, allowing IVT through a 30G needle
- Scalable manufacturing process similar to that of traditional antibody drugs
- Platform-based patented technology, extendable to various targets/indications
- Controllable modulation



Ocular Pharmacokinetics of Intravitreally Injected Protein Therapeutics

Follows first-order elimination kinetics

Intravitreal Injection



Progress in Retinal and Eye Research 57 (2017) 134e185

Route of Elimination

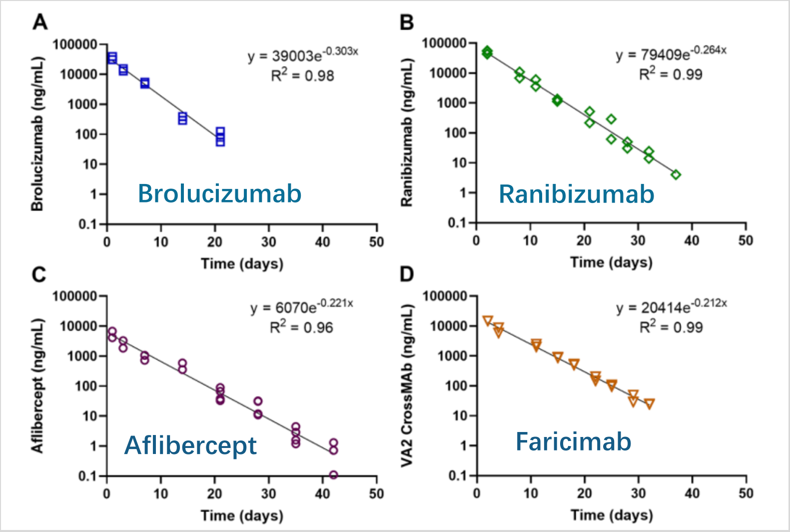
Anterior elimination: the MAJOR ROUTE OF ELIMINATION for biologics drug through **aqueous humor flow**.

Posterior elimination:

- elimination through blood-ocular barriers, permeates through the posterior iris epithelium into iris vein and drained by vortex veins
- through the non-pigmented ciliary epithelium to ciliary muscles and from the ciliary plexus to the episcleral veins
- to the retinal capillaries and through the retinal pigment epithelium (RPE) into the choroid and systemic circulation

Elimination Kinetics

Biologics drug administrated via IVT into eyes follows **FIRST-ORDER ELIMINATION KINETICS**.

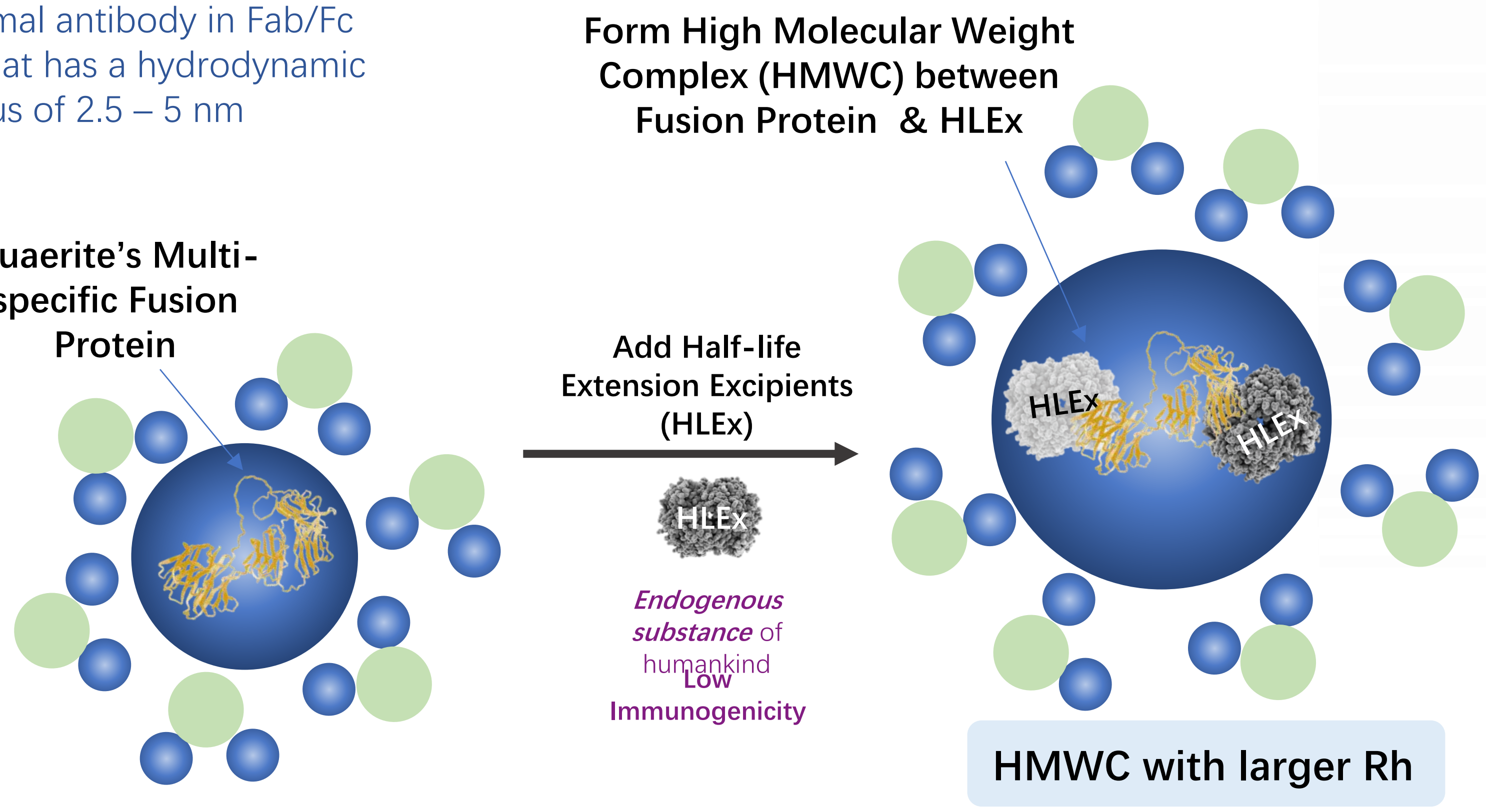


Mol.Pharmaceutics2021,18,2208−2217

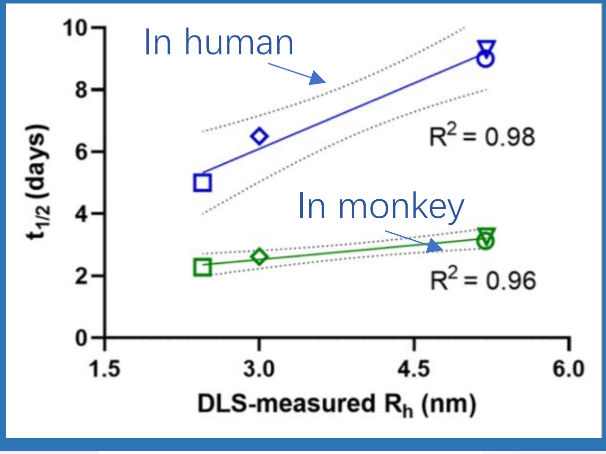
Quaerite's Proprietary PK Optimization Technology Platform (OcuHLE-X®)

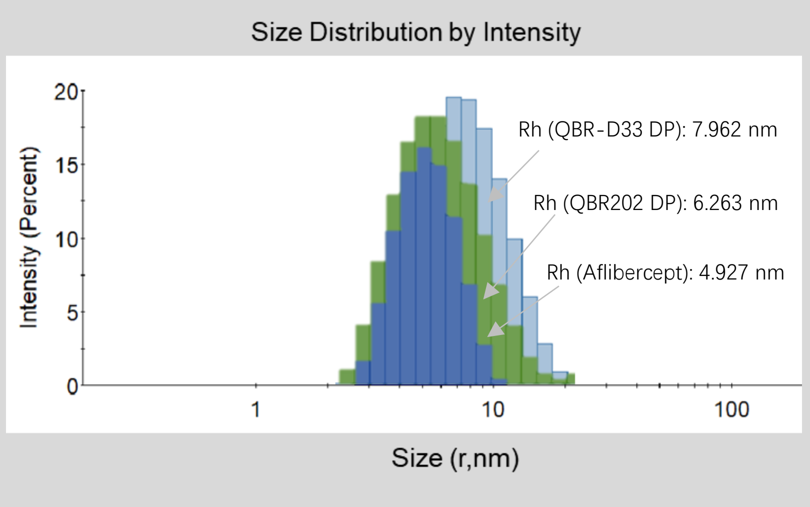
Increase of hydrodynamic radius (Rh) is one of the factors that contribute to half-life extension

Normal antibody in Fab/Fc format has a hydrodynamic radius of 2.5 – 5 nm



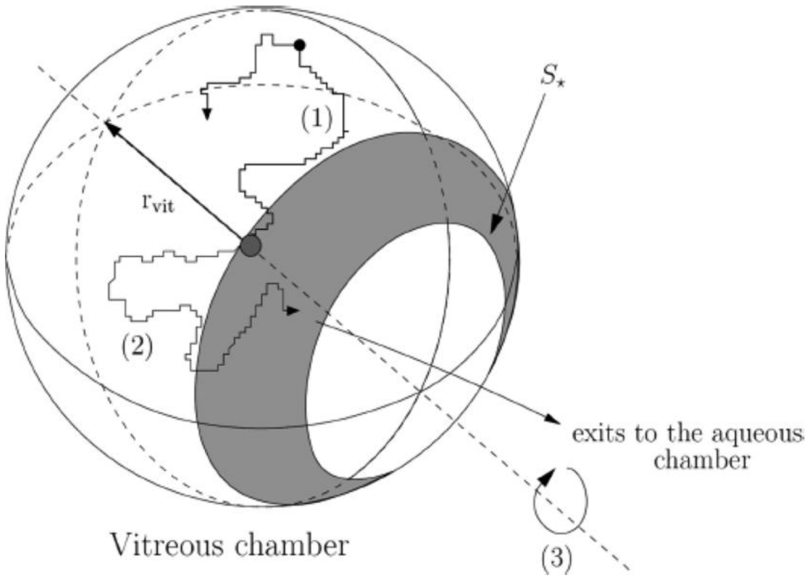
$T_{1/2}$ of biologics drug administrated via IVT is **positively correlated with Rh (nm)**



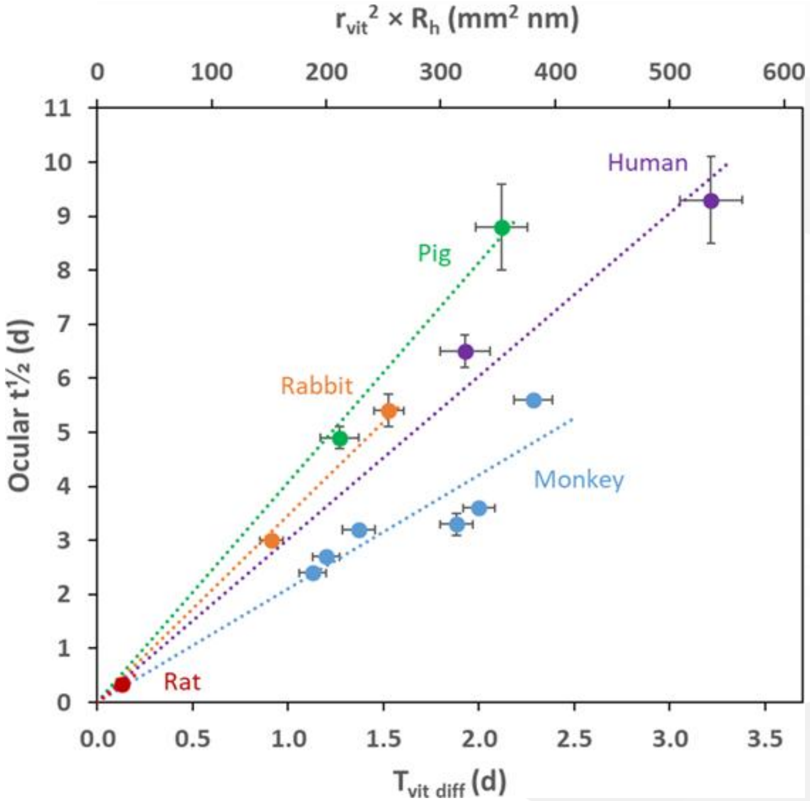


Quaerite's Proprietary PK Optimization Technology Platform (OcuHLE-X®)

Vitreal Diffusion Time ($T_{vit\ diff}$) is positively correlated with hydrodynamic radius (Rh) of a molecule, further explain the longer half-life of OcuHLE-X® molecules



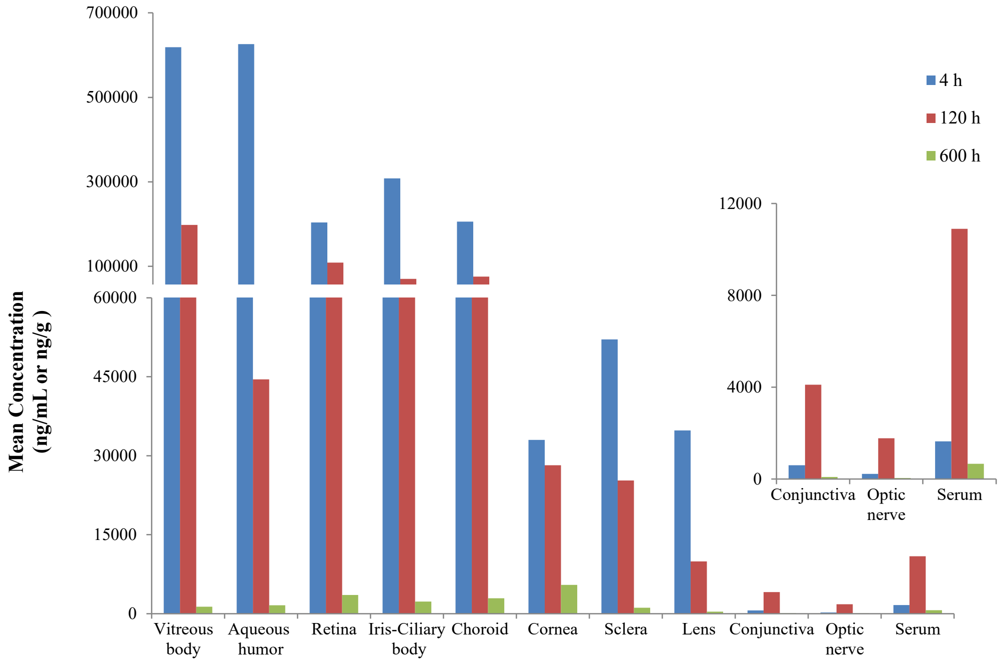
The walk of a molecule that reaches the interface and crosses it into the aqueous chamber



Mol. Pharmaceutics 2020, 17, 695−709

Quaerite's Proprietary PK Optimization Technology Platform (OcuHLE-X®)

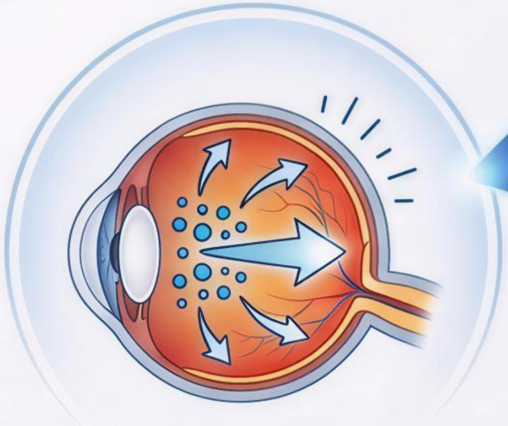
OcuHLE-X helps QBR202 (anti-VEGFA/anti-ANG2 Fusion Protein) achieve favorable ocular tissue distribution in NHP

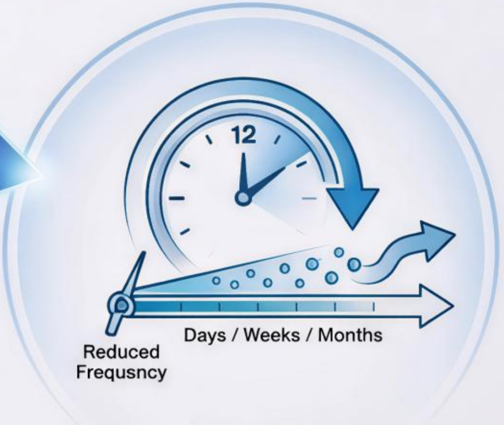


- Exposure levels in the retina and choroid tissues were **46.72%** and **36.48%** of the vitreous humor, respectively.

- Systemic exposure to QBR202 was low, with serum levels representing only 3.60% (AUC_{last}) and 1.76% (C_{max}) of the exposure in the vitreous humor.

OcuHLE-X® Technology Platform



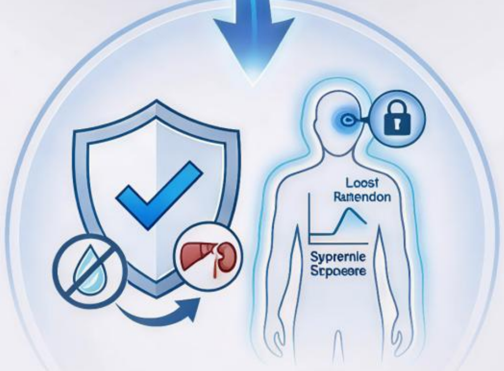


Superior Intraocular Distribution

- Enhanced distribution to retina and choroid
- Higher target tissue exposure

Prolonged Duration of Action

- Extended intraocular half-life
- Reduced injection frequency



Improved Safety Profile

- Low systemic exposure
- Excellent local retention